

# Nathan Shan

## AI Agent Engineer

nathanshanusa@gmail.com · +86 191 9242 2920 · Beijing, China  
nateec.github.io · github.com/nateEc · LinkedIn

## Experience

### AI Agent Engineer

VisionFlow AI

Mar 2026 – Present

Beijing, China

- Designed a user-level skill-learning loop that turns multi-turn agent runs into policy, observation, candidate, and human-confirmation stages with risk tiers, idempotent deduplication, and privacy minimization.
- Built an agent-failure regression platform that converts negative feedback, empty answers, and tool failures into reviewable cases across 6 repositories, 13 domains, and 29 suites; integrated runtime replay, semantic judging, LLM-cost isolation, and incremental Jenkins checks.
- Designed a Skill Router Rule Shadow framework for evaluating candidate rules without affecting user traffic, using stable-field comparison, latency statistics, atomic review bundles, and sensitive-output scanning.
- Built HiPilot Desktop with Electron, the Codex app server, and the t3code workbench, including short-lived model tokens, an SSE proxy, idempotent billing, redacted diagnostics, and macOS release verification.
- Delivered recruiting-agent and AI-interview workflows spanning candidate synchronization, retrieval and matching, outreach reminders, operational reporting, and JD/résumé-driven voice interviews.

### AI Algorithm Development Engineer

Iston AI

May 2025 – Feb 2026

Shanghai, China

- Architected an AI digital-human backend and integrated GPT, RAG, ASR/TTS, and Ollama-powered web search through FastAPI for real-time voice interaction.
- Designed load balancing for Ollama services and moved the pipeline to asynchronous execution, reducing average latency by 70% and increasing throughput by nearly 10×.
- Refactored phoneme parsing and similarity scoring, then built K–12 speaking tasks and dynamic question generation, improving scoring accuracy by approximately 40%.

### Software Engineer Intern

Chinese Historical Christian Database

Jun 2024 – Jul 2024

Boston, MA

- Optimized Neo4j Cypher queries and fixed frontend data-fetching errors, reducing query time by approximately 30% while improving regional church-population data accuracy.

### Software Engineer

BU Spark!

Sep 2023 – Dec 2023

Boston, MA

- Coordinated one designer and three developers to build Petventure with Ionic React, PostgreSQL, and Prisma for pet-friendly venue discovery and distance filtering.

## Selected Projects

### TalkIt Voice Learning Agent

Expo / React Native / Deepgram / Agent OS

Mar 2026 – Apr 2026

- Built a voice-first agent app from the ground up, connecting streaming text, Deepgram STT, sentence-level TTS, quizzes, shadowing, and roleplay; improved long-session continuity through streaming batches, ASR fallback, and Redis session recovery.

### Flipod AI Listening App

Expo / React Native / Cloudflare Workers / D1

Apr 2026

- Modernized the mobile and server stack with word-level captions, CEFR difficulty highlighting, behavior-driven content ranking, structured practice from saved vocabulary, and content localization for 10 native languages.

## Education

### B.S. in Computer Science

Boston University

Sep 2021 – Jan 2025

Boston, MA

Coursework: algorithms, data structures, mobile software engineering, web programming, computer systems, databases, networks, linear algebra, and computational probability.

## Skills

### AI Agents

Skill and tool calling, RAG, LLM applications, agent evaluation, ASR/TTS, model integration, context and session management

### Engineering Operations

Python, TypeScript, JavaScript, FastAPI, React / React Native, Vue, Electron, PostgreSQL, Redis, SQL  
Docker, Nginx, Git, CI/CD, Jenkins, Cloudflare, async systems, observability, automated testing